

$$\text{SS2P2: } \lambda_k(t) = \lambda(t) \cdot p^*(k | t)$$

S2P2 backbone (unchanged)

stacked latent-linear-Hawkes / diagonal SSM, ZOH
LayerNorm'd stack output $u(t)$ feeds BOTH heads

Softmin-bounded rate $\lambda(t)$

$$\begin{aligned} h &= \sigma(W_0 u) \odot \tanh(u) \in (-1, 1)^H \\ z &= c - \text{softplus}(c - w \cdot h - b) \\ \lambda &= s \cdot \text{softplus}(z) \end{aligned}$$

$z \leq c \Rightarrow \text{HARD ceiling } s \cdot \text{softplus}(c)$
(exact thinning bound); floor exactly 0

Rate-neutral marks $p^*(k|t)$

$$p^*(\cdot|t) = \text{softmax}(\text{MLP}(u))$$

on the 62-type simplex

rate-neutral: $\sum_k p^*_k = 1$
 $\Rightarrow \sum_k \lambda_k = \lambda$ regardless of
how nonlinear the mark net is

per-type intensity $\lambda_k(t)$

expressive marks on a provably bounded rate